



MS Excel- Concept Revision Notes

Fomulas | Charts | Tabs | Practical Tools

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23.Tabs

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“In the following pages, we will cover key Excel topics step by step. We start with practical tools and features before moving on to formulas, charts, and tabs. Each topic includes definition, use, and examples to help understand and apply them easily.”

Cell References

1) What is Cell Reference?

- Cell reference refers to the address of a cell used in a formula
- It tells Excel from where to take the value
- Example: A1, B2, C5

2) Types of Cell References

(a) Relative Cell Reference

- Changes automatically when the formula is copied
- Most commonly used reference
- Example:

Formula in C1: =A1+B1

If copied to C2 → =A2+B2

(b) Absolute Cell Reference

- Does not change when the formula is copied
- Dollar sign (\$) is used to lock row and column
- Example:
=A\$1+\$B\$1

(c) Mixed Cell Reference

- Either row or column remains fixed
- Used when one part must stay constant
- Examples:
=A\$1 (Column fixed, row changes)
=A\$1 (Row fixed, column changes)

3) Why Cell References are important

- Makes formulas dynamic and reusable
- Saves time while copying formulas
- Essential for advanced Excel work

4) Practical uses of Cell References

- Tax calculation using fixed tax rate
- Discount or commission calculations
- Financial models and reports

5) Shortcut key (Very important)

- Press **F4** after selecting a cell reference to toggle:
Relative → Absolute → Mixed

6) Common mistakes to avoid

- Forgetting to lock reference when copying formulas
- Using absolute reference unnecessarily
- Wrong combination of row and column locking

Consolidation

1) What is Consolidation?

- Consolidation is used to combine data from multiple worksheets or workbooks into a single summary sheet
- It helps in preparing consolidated reports quickly

2) Why we use Consolidation

- To summarize large data from different sheets
- To prepare combined financial or performance reports
- To save time and avoid manual calculations

3) Types of Consolidation

(a) Consolidation by Position

- Data is consolidated based on the same cell position
- Layout of all sheets must be identical
- Example: Same structure of sales data in all sheets

(b) Consolidation by Category

- Data is consolidated based on matching headings
- Sheet layout can be different
- More flexible and commonly used

4) How to use Consolidation in Excel

- Go to Data Tab → Consolidate
- Select function (Sum, Average, Count, etc.)
- Add references from different sheets/workbooks
- Choose labels (Top row / Left column if required)
- Click OK

5) Functions available in Consolidation

- Sum
- Average
- Count
- Max / Min
- Product

6) Important options

- “Create links to source data” – updates summary automatically when source data changes

7) Practical uses

- Branch-wise sales consolidation
- Monthly expense consolidation
- Department-wise performance reports

8) Limitations of Consolidation

- Source data should be clean and well-structured
- Changes in layout may affect results
- Not suitable for very complex data analysis

Macros

1) What is Macro?

- A Macro is a recorded set of actions that can be repeated automatically
- It is used to automate repetitive tasks in Excel

2) Why we use Macros

- To save time
- To reduce manual work
- To avoid human errors in repetitive tasks

3) Types of Macros

- Recorded Macros – created using Macro Recorder
- VBA Macros – written using Visual Basic for Applications

4) How to create a Macro (Recorded Macro)

- Go to View Tab → Macros → Record Macro
- Give Macro name
- Choose where to store (This Workbook / Personal Macro Workbook)
- Perform required actions
- Stop Recording

5) How to run a Macro

- View Tab → Macros → View Macros → Run
- Or assign Macro to a button or shortcut key

6) Practical uses of Macros

- Formatting reports automatically
- Repetitive data cleaning tasks
- Creating standard reports

7) Important points to remember

- File must be saved as **.xlsm** (Macro-enabled workbook)
- Enable Macros to run them
- Be careful while using Macros from unknown sources

8) Limitations of Macros

- Macros are not supported in Excel Online
- Security risk if used improperly
- Requires VBA knowledge for advanced automation

Flash Fill

1) What is Flash Fill?

- Flash Fill is an Excel feature that automatically fills values when Excel detects a pattern from user input
- It works without using any formula

2) Why we use Flash Fill?

- Saves time in repetitive data entry
- Helps in quick data cleaning and formatting
- Reduces the need for formulas for simple tasks

4) How to use Flash Fill

- Type the required result manually in the adjacent cell
- Select the next cell or range
- Press **Ctrl + E**
- Alternatively, use Data tab → Flash Fill

5) Important points to remember

- Flash Fill works only when the pattern is clear
- It is a one-time action and not dynamic
- It does not update automatically if source data changes
- Available from Excel 2013 onwards

6) Shortcut

- Flash Fill → **Ctrl + E**

7) When NOT to use Flash Fill

- When data changes frequently
- When automatic updating is required

Custom List

1) What is Custom List?

- Custom List is an Excel feature that allows users to create their own predefined lists
- These lists can be used for quick data entry and sorting

2) Why we use Custom List?

- Saves time while entering repetitive data

- Helps maintain consistency in data
- Makes sorting data as per business requirement easy

3) Practical uses of Custom List

- Enter fixed sequences like department names
- Create lists for months in local language or custom order
- Use predefined employee grades or categories
- Sort data in a specific order instead of alphabetical order

4) How to create a Custom List

- Go to File → Options → Advanced
- Scroll to General section
- Click on Edit Custom Lists
- Enter list items manually or import from cells
- Click Add and then OK

5) How to use Custom List

- Type one item from the custom list in a cell
- Drag the fill handle to auto-fill the remaining items
- Use it while sorting data for custom order

6) Important points to remember

- Custom Lists are saved in Excel, not in a single workbook
- They work with the fill handle feature
- Useful for both data entry and sorting

7) When NOT to use Custom List

- When data items change frequently
- When list is required only once

Format Painter

1) What is Format Painter?

- Format Painter is an Excel tool used to copy formatting from one cell to another
- It copies only formatting, not the data or formula

2) Why we use Format Painter?

- To apply same formatting quickly
- To maintain uniform look in worksheets
- To save time compared to manual formatting

3) Practical uses of Format Painter

- Copy font style, size, and colour
- Apply same cell borders and fill colour

- Copy number format like currency or percentage
- Apply same alignment and text formatting

4) How to use Format Painter

- Select the cell with required formatting
- Click on Format Painter
- Select the target cell or range

5) Use on multiple cells

- Double-click on Format Painter
- Select multiple cells or ranges
- Press Esc to stop Format Painter

6) Important points to remember

- Copies formatting only, not values or formulas
- Works for cells, ranges, and even charts
- Formatting is applied instantly

7) Shortcut

- Alt → H → F → P

Clipboard

1) What is Clipboard?

- Clipboard is a temporary storage area in Excel that holds copied or cut items
- It allows users to store and reuse multiple copied items at one time

2) Why we use Clipboard?

- To copy and paste data multiple times
- To manage multiple copied items efficiently
- To save time while working with repetitive data

3) Practical uses of Clipboard

- Copy multiple cells and paste them one by one
- Store frequently used text or values temporarily
- Paste data in different locations without copying again

4) How to open Clipboard pane

- Go to Home tab
- Click the small arrow in the Clipboard group
- Clipboard pane will open on the left side

5) How Clipboard works

- Clipboard can store up to 24 items

- Latest copied item is placed at the top
- Clicking on any item pastes it into the selected cell

6) Important points to remember

- Clipboard clears when Excel is closed
- It works across Office applications
- Items can be removed individually or cleared at once

7) When NOT to rely on Clipboard

- When data needs to be saved permanently
- When working with very sensitive information

Custom Format

1) What is Custom Format?

- Custom Format in Excel is used to change the display of numbers, dates, or text without changing the actual value
- It controls how data appears in a cell

2) Why we use Custom Format?

- To display numbers in required business format
- To improve readability of data
- To keep original values unchanged while changing appearance

3) Practical uses of Custom Format

- Show units like kg, pcs, ₹, % with numbers
- Display leading zeros in codes or IDs
- Show negative numbers in different style
- Format dates in a specific layout

4) How to apply Custom Format

- Select the required cell or range
- Press Ctrl + 1 to open Format Cells
- Go to Number tab
- Select Custom
- Enter the required format code and click OK

5) Common Custom Format examples

- 0.00 → shows two decimal places
- 00000 → displays leading zeros
- #,##0 → applies thousand separator
- dd-mm-yyyy → custom date format

6) Important points to remember

- Custom Format changes display only, not actual data

- Formulas use the original value, not the formatted value
- Same format can be reused in other cells

7) When NOT to use Custom Format

- When actual value needs to be changed
- When data is to be exported in raw numeric form

Wrap Text

1) What is Wrap Text?

- Wrap Text is an Excel feature that displays cell content on multiple lines within the same cell
- It adjusts text according to the column width

2) Why we use Wrap Text?

- To make long text readable without increasing column width
- To keep worksheet layout clean
- To improve presentation of headings and descriptions

3) Practical uses of Wrap Text

- Display long headings in reports
- Show addresses or remarks in one cell
- Present descriptions without merging cells

4) How to apply Wrap Text

- Select the required cell or range
- Go to Home tab
- Click on Wrap Text
- Row height adjusts automatically

5) Important points to remember

- Wrap Text works with row height, not column width
- Manual line break can be added using Alt + Enter
- Wrap Text does not change the actual data

6) Shortcut

- Alt → H → W

7) When NOT to use Wrap Text

- When uniform row height is required
- When exporting data to systems that do not support wrapped text

Merge and Center

1) What is Merge and Center?

- Merge and Center is an Excel feature used to combine multiple selected cells into one cell
- It also centers the content in the merged cell

2) Why we use Merge and Center?

- To create headings across multiple columns
- To improve visual appearance of reports
- To align titles neatly in worksheets

3) Practical uses of Merge and Center

- Create report titles across columns
- Center headings in tables
- Design formatted templates

4) How to apply Merge and Center

- Select the cells to be merged
- Go to Home tab
- Click on Merge and Center

5) Important points to remember

- Only the value in the top-left cell is retained after merging
- Other cell values are removed
- Merged cells can cause issues in sorting and filtering

6) Shortcut

- Alt → H → M → C

7) When NOT to use Merge and Center

- When sorting or filtering data is required
- When working with large datasets

Orientation and Indentation

1) What is Orientation?

- Orientation in Excel is used to change the direction or angle of text within a cell
- It helps display text vertically, diagonally, or at a specific angle

2) What is Indentation?

- Indentation in Excel is used to create space between the cell border and the cell content
- It shifts the text slightly to the right within the cell

3) Why we use Orientation and Indentation?

- To improve readability of data
- To format headings and labels properly
- To align text neatly in reports

4) Practical uses

- Display long column headings vertically using Orientation
- Adjust text position using Indentation for better presentation
- Create structured and clean-looking tables

5) How to apply Orientation

- Select the required cell or range
- Go to Home tab
- Click on Orientation
- Choose the required angle or direction

6) How to apply Indentation

- Select the required cell or range
- Go to Home tab
- Use Increase Indent or Decrease Indent

7) Important points to remember

- Orientation changes text direction, not data
- Indentation affects only alignment, not the cell value
- Both features are for formatting purpose only

Data Validation

1) What is Data Validation?

- Data Validation is an Excel feature used to control what type of data can be entered in a cell
- It helps prevent incorrect or unwanted data entry

2) Why we use Data Validation?

- To ensure data accuracy
- To restrict invalid entries
- To maintain data consistency in worksheets

3) Practical uses of Data Validation

- Create drop-down lists
- Allow only numbers, dates, or text in a cell
- Set minimum and maximum value limits
- Prevent duplicate or incorrect entries

4) How to apply Data Validation

- Select the required cell or range
- Go to Data tab
- Click on Data Validation
- Choose validation criteria and click OK

5) Common Data Validation types

- List → for drop-down values
- Whole Number → restrict integer values
- Decimal → allow decimal numbers
- Date → allow valid date range
- Text Length → restrict number of characters

6) Important points to remember

- Data Validation restricts new entries, not existing data
- Validation rules can be copied to other cells
- Error message appears when invalid data is entered

7) When NOT to rely only on Data Validation

- When data is pasted from external sources without checking
- When data cleaning is required for existing data

Conditional Formatting

1) What is Conditional Formatting?

- Conditional Formatting is an Excel feature used to format cells automatically based on specific conditions
- It changes cell colour, font, or style when criteria are met

2) Why we use Conditional Formatting?

- To highlight important data quickly
- To identify trends, outliers, or errors
- To improve readability and visual analysis

3) Practical uses of Conditional Formatting

- Highlight cells greater or less than a value
- Colour-scale to show data progression
- Data bars to compare values visually
- Highlight duplicate or unique values
- Apply custom formatting using formulas

4) How to apply Conditional Formatting

- Select the required cell or range
- Go to Home tab → Conditional Formatting
- Choose a rule type and set criteria
- Click OK to apply

5) Important points to remember

- Conditional Formatting is visual only; it does not change cell value
- Multiple rules can be applied to a range
- Rule priority can be managed using “Manage Rules” option
- Can slow down Excel if applied to very large datasets

6) When NOT to rely only on Conditional Formatting

- When actual value changes are required
- When formatting is needed for printing purposes

Filter and Advanced Filter

1) What is Filter?

- Filter is an Excel feature used to display only the rows that meet specific criteria
- It hides the rows that do not match the condition

2) What is Advanced Filter?

- Advanced Filter allows more complex filtering using multiple criteria and locations
- Can filter data in place or copy filtered data to another location

3) Why we use Filter and Advanced Filter?

- To analyze large datasets efficiently
- To view only relevant data
- To extract data based on multiple conditions

4) Practical uses of Filter

- Display records of a specific department or category
- Show sales greater than a certain value
- Filter data by dates, numbers, or text
- Extract filtered data to a new location using Advanced Filter

5) How to apply Filter

- Select the data range
- Go to Data tab → Filter
- Click drop-down arrows in column headers
- Select criteria to filter data

6) How to apply Advanced Filter

- Select the data range
- Go to Data tab → Advanced
- Choose “Filter in place” or “Copy to another location”
- Define criteria range and click OK

7) Important points to remember

- Filter hides data, it does not delete it
- Advanced Filter can use multiple criteria and formulas
- Always include headers in the data range

8) When NOT to use Filter / Advanced Filter

- When data needs permanent deletion
- When complex dynamic filtering is required repeatedly (Pivot Tables are better)

What-If Analysis

1) What is What-If Analysis?

- A feature in Excel to see the effect of changing input values on formulas or results
- Helps in decision-making by simulating different scenarios

2) Why we use What-If Analysis?

- Predict outcomes based on different assumptions
- Compare multiple scenarios quickly
- Plan and make informed business decisions

3) Main Tools of What-If Analysis

a) Scenario Manager

- Allows creating and saving multiple sets of input values (scenarios)
- Useful to compare different outcomes in one report
- Example: Profit calculation for best case, worst case, and expected case

b) Goal Seek

- Finds the input value required to achieve a specific result
- Works with one input cell and one output cell
- Example: Find required sales quantity to achieve target profit

c) Data Table

- Shows multiple results for one or two varying input values
- Can be one-variable or two-variable table
- Example: Loan EMI for different interest rates and tenures

4) How to use What-If Analysis

- Go to Data tab → What-If Analysis
- Choose Scenario Manager, Goal Seek, or Data Table
- Enter input cells and criteria
- Apply and analyze results

5) Important points to remember

- Original data remains unchanged
- Helps in planning, forecasting, and comparing alternatives
- Scenarios and tables can be saved for future use

6) When NOT to rely only on What-If Analysis

- When actual values are required
- When dynamic calculations are needed continuously (use formulas/PivotTables)

Text to Column

1) What is Text to Columns?

- Text to Columns is an Excel feature used to split the content of one cell into multiple cells
- Splitting is done based on a delimiter or fixed width

2) Why we use Text to Columns?

- To separate combined data like full name, address, or codes
- To clean imported data efficiently
- To organize data for analysis or reporting

3) Practical uses of Text to Columns

- Split full name into first name and last name
- Separate address into street, city, and state
- Split codes like “ABC-123” into “ABC” and “123”
- Convert dates or numbers stored as text into separate cells

4) How to use Text to Columns

- Select the column to split
- Go to Data tab → Text to Columns
- Choose Delimited (e.g., comma, space) or Fixed Width
- Set delimiters or column breaks
- Click Finish to split data

5) Important points to remember

- Original data is replaced by split data (backup recommended)
- Works on one column at a time
- Can be combined with Flash Fill for automation

6) When NOT to use Text to Columns

- When data structure is irregular or inconsistent
- When splitting is needed dynamically (formulas or Flash Fill are better)

Pivot Table

1) What is Pivot Table?

- Pivot Table is an Excel feature used to summarize, analyze, and explore large datasets
- It allows dynamic rearrangement of rows, columns, and values

2) Why we use Pivot Table?

- To quickly summarize large datasets
- To analyze trends, totals, and comparisons
- To create reports without formulas or manual calculations

3) Practical uses of Pivot Table

- Summarize sales by product, region, or salesperson
- Calculate total, average, minimum, or maximum values

- Count number of orders or customers
- Filter and group data dynamically for reporting

4) How to create a Pivot Table

- Select the data range
- Go to Insert tab → Pivot Table
- Choose location (new worksheet or existing worksheet)
- Drag fields into Rows, Columns, Values, and Filters areas
- Apply formatting and filters as needed

5) Important points to remember

- Pivot Table works on data, not on formulas directly
- Source data should have headers and no blank rows/columns
- Can be refreshed after source data changes
- Multiple calculations like Sum, Average, Count can be applied

6) When NOT to rely only on Pivot Table

- When data needs to be edited directly in the table
- When extremely complex calculations with multiple dependencies are required

Sparklines

1) What is Sparklines?

- Sparklines are small, simple charts placed inside a single cell to show trends or patterns
- They provide a quick visual representation of data

2) Why we use Sparklines?

- To quickly analyze trends in data
- To show increase, decrease, or patterns visually
- To save space compared to full-size charts

3) Practical uses of Sparklines

- Show monthly sales trend in one row
- Visualize stock price movements
- Compare performance across products or departments
- Highlight peaks, lows, and overall trends

4) How to insert Sparklines

- Select the data range
- Go to Insert tab → Sparklines (Line, Column, or Win/Loss)
- Choose location to place Sparklines
- Click OK to display

5) Important points to remember

- Sparklines are for visualization only; they do not change the original data

- Can be customized with colours, markers, and styles
- Works best for small datasets

6) When NOT to use Sparklines

- When detailed charts with axes, legends, or multiple series are needed
- When analyzing very large datasets

Series

1) What is Series?

- Series in Excel is a feature used to fill a range of cells with sequential numbers, dates, or other patterns automatically
- It saves time compared to manual entry

2) Why we use Series?

- To quickly generate numbers, dates, or text patterns
- To maintain consistency in sequential data
- To avoid manual errors in large datasets

3) Practical uses of Series

- Fill serial numbers in rows or columns
- Generate dates for a month or year
- Create repeating or incremental patterns
- Auto-fill weekdays, months, or custom lists

4) How to use Series

- Select starting cell
- Go to Home tab → Fill → Series
- Choose series type (Linear, Growth, Date, AutoFill)
- Set step value and stop value, then click OK

5) Important points to remember

- Series can be horizontal or vertical
- Step value determines increment
- Works with numbers, dates, and custom lists

6) When NOT to use Series

- When irregular patterns are required
- When data is not sequential or needs manual input

Text Function

1) LOWER

- Converts all letters in a text string to lowercase

- Syntax: =LOWER(text)
- Example: =LOWER("HELLO") → hello

2) UPPER

- Converts all letters in a text string to uppercase
- Syntax: =UPPER(text)
- Example: =UPPER("hello") → HELLO

3) PROPER

- Capitalizes the first letter of each word
- Syntax: =PROPER(text)
- Example: =PROPER("hello world") → Hello World

4) TRIM

- Removes all extra spaces from text except single spaces between words
- Syntax: =TRIM(text)
- Example: =TRIM(" hello world ") → hello world

5) LEN

- Returns the total number of characters in a text string (including spaces)
- Syntax: =LEN(text)
- Example: =LEN("Excel 2026") → 10

6) LEFT

- Returns the first N characters from the left of a text string
- Syntax: =LEFT(text, num_chars)
- Example: =LEFT("Excel", 3) → Exc

7) RIGHT

- Returns the last N characters from the right of a text string
- Syntax: =RIGHT(text, num_chars)
- Example: =RIGHT("Excel", 2) → el

8) MID

- Returns N characters from the middle of a text string starting at a specified position
- Syntax: =MID(text, start_num, num_chars)
- Example: =MID("Excel", 2, 3) → xce

9) CONCATENATE

- Joins two or more text strings into one
- Syntax (old): =CONCATENATE(text1, text2, ...)
- Syntax (new): =CONCAT(text1, text2, ...)
- Example: =CONCAT("Hello", " ", "World") → Hello World

10) Important points to remember

- All functions work on text values; numbers may need conversion to text

- CONCATENATE is replaced by CONCAT in newer Excel versions
- MID, LEFT, RIGHT depend on character positions

11) When NOT to use these functions

- When numeric calculation is required

"We have now covered important Excel features and text functions. Next, we will explore commonly used Excel formulas step-by-step, which are very useful for practical work."

Formulas & Functions

Sum

1) What is SUM?

- SUM is an Excel formula used to add numbers in a range or individual cells
- Syntax: =SUM(number1, number2, ...) OR =SUM(range)

2) Why we use SUM?

- To quickly calculate total of multiple numbers
- To save time compared to manual addition
- Works for continuous and non-continuous ranges

3) Practical uses of SUM

- Add monthly sales figures
- Total expenses or incomes
- Sum selected cells from different rows/columns

4) How to use SUM

- Example 1 (Range): =SUM(A1:A5) → Adds values from A1 to A5
- Example 2 (Multiple cells): =SUM(A1, B3, C5) → Adds values in selected cells
- Can be combined with other functions: =SUM(A1:A5, C1:C5)

5) Important points to remember

- SUM ignores empty cells and text values
- Can be used with rows, columns, and mixed references
- AutoSum button (Σ) is a quick alternative

6) When NOT to rely only on SUM

- When conditional addition is required (use SUMIF or SUMIFS)
- When dynamic filtering is needed (use SUBTOTAL)

SumIf

1) What is SUMIF?

- SUMIF is an Excel formula used to add values in a range that meet a specific condition
- Syntax: =SUMIF(range, criteria, [sum_range])

2) Why we use SUMIF?

- To calculate total based on a condition
- To analyze data without manually filtering
- Saves time and reduces errors in conditional summing

3) Practical uses of SUMIF

- Add sales of a specific product
- Sum expenses greater than a certain amount
- Total scores of a particular student or category

4) How to use SUMIF

- Example 1: =SUMIF(A2:A10, "Apple", B2:B10) → Adds values in B2:B10 where A2:A10 is "Apple"
- Example 2: =SUMIF(C2:C10, ">1000") → Adds values in C2:C10 greater than 1000
- Can combine text, numbers, and logical operators in criteria

5) Important points to remember

- SUMIF allows only one condition; for multiple conditions, use SUMIFS
- Criteria can be text, number, or expression
- Empty cells in sum range are ignored

6) When NOT to rely only on SUMIF

- When multiple criteria are needed (use SUMIFS)
- When complex logic is required across multiple ranges

Average

1) What is AVERAGE?

- AVERAGE is an Excel formula used to calculate the arithmetic mean of numbers in a range or selected cells
- Syntax: =AVERAGE(number1, number2, ...) Or =AVERAGE(range)

2) Why we use AVERAGE?

- To quickly find the mean value of a dataset
- To analyze trends and performance
- Saves time compared to manual calculation

3) Practical uses of AVERAGE

- Calculate average sales or revenue
- Determine average marks of students
- Find average expense per month or category

4) How to use AVERAGE

- Example 1 (Range): =AVERAGE(A1:A10) → Calculates average of values from A1 to A10
- Example 2 (Multiple cells): =AVERAGE(A1, B3, C5) → Calculates average of selected cells
- Can be combined with other functions: =AVERAGE(A1:A10, C1:C5)

5) Important points to remember

- AVERAGE ignores empty cells and text values
- Can be used with rows, columns, or mixed references
- Use AVERAGEIF/AVERAGEIFS for conditional averaging

6) When NOT to rely only on AVERAGE

- When certain values need to be excluded based on conditions (use AVERAGEIF/AVERAGEIFS)
- When weighted average is required (use SUMPRODUCT or manual formula)

Min

1) What is MIN?

- MIN is an Excel formula used to find the smallest number in a range or set of selected cells
- Syntax: =MIN(number1, number2, ...) OR =MIN(range)

2) Why we use MIN?

- To quickly identify the lowest value in a dataset
- To analyze minimum performance, cost, or value
- Saves time compared to manually scanning data

3) Practical uses of MIN

- Find the minimum sales, expense, or score
- Identify the earliest date or lowest number in records
- Compare performance across multiple categories

4) How to use MIN

- Example 1 (Range): =MIN(A1:A10) → Finds the smallest value in A1 to A10
- Example 2 (Multiple cells): =MIN(A1, B3, C5) → Finds the smallest value among selected cells
- Can be combined with other functions for analysis: =MIN(A1:A10, C1:C5)

5) Important points to remember

- MIN ignores empty cells and text values
- Works with numbers, dates, and times
- Use MINIFS for conditional minimum calculation

6) When NOT to rely only on MIN

- When conditional minimum is needed (use MINIFS)
- When a weighted minimum or specific subset analysis is required

Max

1) What is MAX?

- MAX is an Excel formula used to find the largest number in a range or set of selected cells
- Syntax: =MAX(number1, number2, ...) OR =MAX(range)

2) Why we use MAX?

- To quickly identify the highest value in a dataset
- To analyze maximum performance, cost, or value
- Saves time compared to manually scanning data

3) Practical uses of MAX

- Find the maximum sales, expense, or score
- Identify the latest date or highest number in records
- Compare performance across multiple categories

4) How to use MAX

- Example 1 (Range): =MAX(A1:A10) → Finds the largest value in A1 to A10
- Example 2 (Multiple cells): =MAX(A1, B3, C5) → Finds the largest value among selected cells
- Can be combined with other functions for analysis: =MAX(A1:A10, C1:C5)

5) Important points to remember

- MAX ignores empty cells and text values
- Works with numbers, dates, and times
- Use MAXIFS for conditional maximum calculation

6) When NOT to rely only on MAX

- When conditional maximum is needed (use MAXIFS)
- When a weighted maximum or specific subset analysis is required

Count

1) What is COUNT?

- COUNT is an Excel formula used to count the number of numeric values in a range or selected cells
- Syntax: =COUNT(value1, value2, ...) or =COUNT(range)

2) Why we use COUNT?

- To quickly determine how many numeric entries are present
- Helps in data analysis and reporting
- Saves time compared to manual counting

3) Practical uses of COUNT

- Count the number of sales entries in a list
- Count the number of employees with numeric ID values
- Track the number of numeric responses in surveys

4) How to use COUNT

- Example 1 (Range): =COUNT(A1:A10) → Counts numeric values from A1 to A10
- Example 2 (Multiple cells): =COUNT(A1, B3, C5) → Counts numeric values among selected cells
- Works with dates and times since Excel stores them as numbers

5) Important points to remember

- COUNT ignores empty cells and text values
- To count all entries including text, use COUNTA
- For conditional counting, use COUNTIF/COUNTIFS

6) When NOT to rely only on COUNT

- When you need to count text entries (use COUNTA)
- When you need to count based on conditions (use COUNTIF/COUNTIFS)

CountA

1) What is COUNTA?

- COUNTA is an Excel formula used to count all non-empty cells in a range, including numbers, text, logical values, and errors
- Syntax: =COUNTA(value1, value2, ...) Or =COUNTA(range)

2) Why we use COUNTA?

- To quickly determine how many cells contain any data
- Useful for tracking responses, entries, or filled cells
- Helps in reporting and data completeness checks

3) Practical uses of COUNTA

- Count the number of students who submitted assignments
- Count filled cells in a survey or form
- Determine the total number of entries in a list

4) How to use COUNTA

- Example 1 (Range): =COUNTA(A1:A10) → Counts all non-empty cells in A1 to A10
- Example 2 (Multiple cells): =COUNTA(A1, B3, C5) → Counts all non-empty selected cells
- Works with numbers, text, logical values (TRUE/FALSE), and errors

5) Important points to remember

- COUNTA counts all non-empty cells, including text, numbers, logical values, and errors
- Empty cells are ignored
- For counting only numbers, use COUNT

6) When NOT to rely only on COUNTA

- When only numeric values need to be counted (use COUNT)
- When conditional counting is required (use COUNTIF/COUNTIFS)

CountBlank

1) What is COUNTBLANK?

- COUNTBLANK is an Excel formula used to count the number of empty cells in a range
- Syntax: =COUNTBLANK(range)

2) Why we use COUNTBLANK?

- To quickly identify missing or empty data in a dataset
- Helps in data validation and completeness checks
- Useful for reporting and analysis

3) Practical uses of COUNTBLANK

- Count unfilled survey responses
- Identify missing sales or attendance entries
- Check empty cells in a list or table before analysis

4) How to use COUNTBLANK

- Example 1 (Range): =COUNTBLANK(A1:A10) → Counts empty cells from A1 to A10
- Example 2 (Multiple ranges, combined with SUM): =COUNTBLANK(A1:A10)+COUNTBLANK(B1:B10) → Counts empty cells in both ranges

5) Important points to remember

- Only truly empty cells are counted
- Cells with formulas returning empty string ("") are counted as empty
- Text or numeric entries are ignored

6) When NOT to rely only on COUNTBLANK

- When you want to count only numeric blanks (use COUNTIF with criteria = "")
- When conditional blanks based on other criteria are required (use COUNTIFS)

CountIf and CountIfs

1) What is COUNTIF?

- COUNTIF is an Excel formula used to count the number of cells in a range that meet a single condition
- Syntax: =COUNTIF(range, criteria)

2) What is COUNTIFS?

- COUNTIFS is an Excel formula used to count the number of cells that meet multiple conditions across one or more ranges
- Syntax: =COUNTIFS(criteria_range1, criteria1, [criteria_range2, criteria2], ...)

3) Why we use COUNTIF/COUNTIFS?

- To quickly count cells based on specific conditions

- Useful for data analysis, reporting, and quality checks
- Saves time compared to manual counting

4) Practical uses of COUNTIF/COUNTIFS

- COUNTIF: Count the number of sales above 1000 → =COUNTIF(B2:B20, ">1000")
- COUNTIF: Count the number of "Apple" entries → =COUNTIF(A2:A20, "Apple")
- COUNTIFS: Count sales above 1000 in a specific region → =COUNTIFS(B2:B20, ">1000", C2:C20, "North")
- COUNTIFS: Count students who scored >70 in Math and >75 in Science

5) Important points to remember

- COUNTIF allows only one condition
- COUNTIFS allows multiple conditions with AND logic
- Criteria can be numbers, text, expressions, or logical operators
- Wildcards (* for any characters, ? for single character) can be used in text criteria

6) When NOT to rely only on COUNTIF/COUNTIFS

- When OR logic is needed across multiple criteria (use SUMPRODUCT or helper columns)
- When counting based on complex dynamic conditions not supported by COUNTIFS

If

1) What is IF?

- IF is an Excel formula used to perform a logical test and return one value if the condition is TRUE and another if FALSE
- Syntax: =IF(logical_test, value_if_true, value_if_false)

2) Why we use IF?

- To make decisions based on conditions in data
- Helps automate analysis and reporting
- Saves time compared to manual checks

3) Practical uses of IF

- Assign "Pass" or "Fail" based on score → =IF(B2>=50, "Pass", "Fail")
- Give bonus if sales exceed a target → =IF(C2>1000, 500, 0)
- Highlight high/low values in dashboards

4) How to use IF

- Example 1: =IF(A1>10, "Yes", "No") → Returns "Yes" if A1>10, else "No"
- Example 2 (Nested IF): =IF(B2>=90, "A", IF(B2>=75, "B", "C")) → Assign grades based on score

5) Important points to remember

- IF can be nested for multiple conditions
- Returns exactly one of the two values based on condition
- Logical test can include numbers, text, dates, or expressions

6) When NOT to rely only on IF

- For multiple complex conditions, nested IF can become hard to manage (use IFS in newer Excel or SWITCH)
- When array-based or dynamic calculations are needed, consider combining with other functions

IfError

1) What is IFERROR?

- IFERROR is an Excel formula used to catch and handle errors in formulas
- Returns a custom value if an error occurs, otherwise returns the formula result
- Syntax: =IFERROR(value, value_if_error)

2) Why we use IFERROR?

- To prevent error messages like #DIV/0!, #N/A, or #VALUE! from appearing
- To make reports cleaner and more readable
- Useful when working with VLOOKUP, division, or other formulas that may generate errors

3) Practical uses of IFERROR

- Handle division by zero → =IFERROR(A2/B2, 0) → Returns 0 instead of #DIV/0!
- Handle missing lookup values → =IFERROR(VLOOKUP(101,A2:B10,2,FALSE),"Not Found")
- Replace formula errors with blank → =IFERROR(C2*D2,"")

4) How to use IFERROR

- Example 1: =IFERROR(A1/B1, "Error") → Returns "Error" if division fails
- Example 2: =IFERROR(VLOOKUP(E2,A2:B10,2,FALSE),0) → Returns 0 if value not found

5) Important points to remember

- IFERROR works for all types of errors: #DIV/0!, #N/A, #VALUE!, #REF!, etc.
- Helps in creating clean dashboards and reports
- Use cautiously; hiding errors without analysis may mask real problems

6) When NOT to rely only on IFERROR

- When the cause of the error needs to be identified and corrected
- When error masking may lead to wrong business decisions

NestedIf

1) What is Nested IF?

- Nested IF is an Excel formula where one IF function is placed inside another IF to test multiple conditions
- Syntax: =IF(condition1, value_if_true1, IF(condition2, value_if_true2, value_if_false2))

2) Why we use Nested IF?

- To evaluate multiple conditions in a single formula

- Helps automate decision-making with more than two outcomes
- Saves time compared to using multiple separate IF formulas

3) Practical uses of Nested IF

- Assign grades based on marks → =IF(A1>=90,"A",IF(A1>=75,"B",IF(A1>=50,"C","F")))
- Categorize sales → High, Medium, Low based on thresholds
- Determine bonus tiers based on performance

4) How to use Nested IF

- Example 1: =IF(B2>1000,"Excellent",IF(B2>500,"Good","Average")) → Returns different text based on value
- Example 2: =IF(C2="Red",10,IF(C2="Blue",20,30)) → Assigns numeric values based on text

5) Important points to remember

- Each IF tests one condition and returns a value or another IF
- Too many nested IFs (more than 7-8) can become complex and hard to manage
- In newer Excel, IFS function is preferred for multiple conditions

6) When NOT to rely only on Nested IF

- When multiple conditions are numerous, use IFS or SWITCH for clarity
- When formulas need to be dynamic or easier to maintain

Today

1) What is TODAY?

- TODAY is an Excel formula used to return the current date based on your system date
- Syntax: =TODAY()
- Does not require any arguments

2) Why we use TODAY?

- To display the current date dynamically
- Useful in date calculations, reports, dashboards, and age or tenure calculations
- Updates automatically whenever the file is opened or recalculated

3) Practical uses of TODAY

- Calculate age or service period → =TODAY()-B2 (if B2 has birth/joining date)
- Highlight overdue tasks → =IF(A2<TODAY(),"Overdue","Pending")
- Track current date in dashboards and reports

4) How to use TODAY

- Example 1: =TODAY() → Returns current date, e.g., 21-Jan-2026
- Example 2 (with calculation): =DATEDIF(B2,TODAY(),"Y") → Calculates age in years from birth date in B2

5) Important points to remember

- TODAY returns only the date, not time

- Updates automatically each day
- Can be combined with other formulas for dynamic date calculations

6) When NOT to rely only on TODAY

- When a fixed/static date is required (use manual entry or copy-paste as value)
- When historical snapshots of reports are needed

Now

1) What is NOW?

- NOW is an Excel formula used to return the current date and time based on your system clock
- Syntax: =NOW()
- Does not require any arguments

2) Why we use NOW?

- To display the current date and time dynamically
- Useful in timestamping, dashboards, and real-time calculations
- Updates automatically whenever the file is opened or recalculated

3) Practical uses of NOW

- Display current date and time in reports or forms
- Track timestamp of transactions or actions → =IF(A2<NOW(),"Overdue","Pending")
- Combine with other formulas for age, tenure, or elapsed time calculations

4) How to use NOW

- Example 1: =NOW() → Returns current date & time, e.g., 21-Jan-2026 14:30
- Example 2 (with calculation): =NOW()-B2 → Calculates time elapsed from date-time in B2

5) Important points to remember

- NOW returns both date and time
- Updates automatically each time Excel recalculates
- Can be formatted to show only date or only time using cell formatting

6) When NOT to rely only on NOW

- When a fixed/static timestamp is required (use manual entry or copy-paste as value)
- When historical snapshots of reports are needed

DateDif

1) What is DATEDIF?

- DATEDIF is an Excel formula used to calculate the difference between two dates in days, months, or years
- Syntax: =DATEDIF(start_date, end_date, unit)

2) Why we use DATEDIF?

- To calculate age, tenure, or duration between two dates
- Useful in HR, finance, and reporting
- Helps in precise date difference calculation without manual work

3) Practical uses of DATEDIF

- Calculate age → =DATEDIF(B2,TODAY(),"Y") (B2 = Birthdate)
- Calculate service period → =DATEDIF(C2,D2,"M") → Months of service between C2 and D2
- Track project duration → =DATEDIF(StartDate, EndDate, "D") → Total days

4) How to use DATEDIF

- Units:
 - "Y" → Complete years between dates
 - "M" → Complete months between dates
 - "D" → Total days between dates
 - "MD" → Difference in days ignoring months and years
 - "YM" → Difference in months ignoring years
 - "YD" → Difference in days ignoring years
 - Example: =DATEDIF(A2,B2,"YD") → Days difference ignoring years

5) Important points to remember

- Start date must be earlier than end date
- Returns numeric values
- Works well with TODAY() for dynamic calculations

6) When NOT to rely only on DATEDIF

- When you need time difference including hours/minutes (use NOW() or subtraction)
- Inconsistent results if start_date > end_date

Edate

1) What is EDATE?

- EDATE is an Excel formula used to calculate a date that is a specified number of months before or after a given date
- Syntax: =EDATE(start_date, months)

2) Why we use EDATE?

- To calculate due dates, maturity dates, or expiry dates
- Useful in finance, HR, and project planning
- Helps in automating recurring date calculations

3) Practical uses of EDATE

- Calculate next month's payment date → =EDATE(A2,1) (A2 = Start Date)

- Calculate maturity date 6 months ahead → =EDATE(B2,6)
- Calculate previous month → =EDATE(C2,-1)

4) How to use EDATE

- Example 1: =EDATE("21-Jan-2026",3) → Returns 21-Apr-2026
- Example 2 (cell reference): =EDATE(A2, -6) → 6 months before date in A2

5) Important points to remember

- Positive months → future date, negative months → past date
- Returns a valid date that can be formatted
- Works well with TODAY() for dynamic calculations

6) When NOT to rely only on EDATE

- When you need exact days difference including partial months (use DATEDIF or simple subtraction)
- When dealing with time (hours/minutes), EDATE only handles dates

Eomonth

1) What is EOMONTH?

- EOMONTH is an Excel formula used to return the last day of the month, a specified number of months before or after a given date
- Syntax: =EOMONTH(start_date, months)

2) Why we use EOMONTH?

- To calculate month-end dates for payments, salaries, or reporting
- Useful in finance, accounting, and project planning
- Helps automate recurring month-end calculations

3) Practical uses of EOMONTH

- Find month-end date for next month → =EOMONTH(A2,1)
- Find month-end date 3 months ago → =EOMONTH(B2,-3)
- Automate due dates or maturity dates that fall at month-end

4) How to use EOMONTH

- Example 1: =EOMONTH("21-Jan-2026",1) → Returns 28-Feb-2026
- Example 2 (cell reference): =EOMONTH(A2,0) → Returns last date of the month of A2
- Can be combined with DAY, TEXT, or other formulas for reporting

5) Important points to remember

- Positive months → future month-end, negative months → past month-end
- Returns a valid date that can be formatted
- Often used with TODAY() for dynamic month-end calculations

6) When NOT to rely only on EOMONTH

- When exact day difference is required (use DATEDIF or simple subtraction)
- When you need a date that is not month-end, EOMONTH is not suitable

Networkdays

1) What is NETWORKDAYS?

- NETWORKDAYS is an Excel formula used to calculate the number of working days (excluding weekends) between two dates
- Syntax: =NETWORKDAYS(start_date, end_date, [holidays])

2) Why we use NETWORKDAYS?

- To calculate project durations, employee leave, or task timelines
- Useful in HR, finance, and project management
- Automatically excludes weekends and optional holidays

3) Practical uses of NETWORKDAYS

- Calculate total working days between joining date and today → =NETWORKDAYS(B2,TODAY())
- Project timeline calculation → =NETWORKDAYS(StartDate, EndDate, HolidaysRange)
- Payroll or leave calculations excluding weekends and holidays

4) How to use NETWORKDAYS

- Example 1: =NETWORKDAYS("01-Jan-2026","21-Jan-2026") → Returns 15 (weekends excluded)
- Example 2 (with holidays): =NETWORKDAYS(A2,B2,D2:D5) → Counts working days excluding dates in D2:D5

5) Important points to remember

- WEEKENDS are Saturday & Sunday by default
- Holidays are optional but can be provided as a range
- Returns numeric value (number of working days)

6) When NOT to rely only on NETWORKDAYS

- If custom weekends are needed (use NETWORKDAYS.INTL)
- For exact total days including weekends, use simple subtraction

Networkdays.intl

1) What is NETWORKDAYS.INTL?

- NETWORKDAYS.INTL is an Excel formula used to calculate the number of working days between two dates, with custom weekends
- Syntax: =NETWORKDAYS.INTL(start_date, end_date, [weekend], [holidays])

2) Why we use NETWORKDAYS.INTL?

- To calculate working days with flexible weekend settings (e.g., Friday-Saturday weekend)

- Useful in international projects, HR, and payroll calculations
- Allows exclusion of holidays in addition to custom weekends

3) Practical uses of NETWORKDAYS.INTL

- Calculate working days for employees in countries with non-standard weekends →
=NETWORKDAYS.INTL(A2,B2,7) (Sunday only weekend)
- Project timeline with custom weekends and holidays → =NETWORKDAYS.INTL(StartDate, EndDate, 2, HolidayRange)
- Payroll and leave management excluding holidays and specific weekends

4) How to use NETWORKDAYS.INTL

- Weekend argument can be a number (1–17) or a 7-character string:
 - Number: 1 = Sat-Sun, 2 = Sun-Mon, etc.
 - String: "0000011" → 1=Weekend, 0=Working day (Sat & Sun weekend here)
 - Example 1: =NETWORKDAYS.INTL("01-Jan-2026","21-Jan-2026",1) → Returns working days excluding standard Sat-Sun
 - Example 2 (with holidays): =NETWORKDAYS.INTL(A2,B2,"0000011",D2:D5) → Custom weekend and holiday range

5) Important points to remember

- Provides flexibility for global projects and non-standard weekends
- Holidays are optional but recommended for accurate calculation
- Returns numeric value (number of working days)

6) When NOT to rely only on NETWORKDAYS.INTL

- For simple standard Sat-Sun weekends, NETWORKDAYS is sufficient
- For calculations including weekends, use simple subtraction

Or & And

1) What is OR?

- OR is an Excel logical function used to test multiple conditions and returns TRUE if **any one** of the conditions is TRUE
- Syntax: =OR(condition1, condition2, ...)

2) What is AND?

- AND is an Excel logical function used to test multiple conditions and returns TRUE only if **all** conditions are TRUE
- Syntax: =AND(condition1, condition2, ...)

3) Why we use OR & AND?

- To perform logical tests in formulas and automate decisions
- Often combined with IF for complex conditional outputs
- Saves time and avoids manual checks

4) Practical uses of OR & AND

- OR Example: =IF(OR(A2>100,B2>200),"Bonus","No Bonus") → Returns “Bonus” if either condition is TRUE
- AND Example: =IF(AND(A2>100,B2>200),"Bonus","No Bonus") → Returns “Bonus” only if both conditions are TRUE
- Combined in nested logical tests → =IF(AND(A2>50, OR(B2>100,C2>200)),"Pass","Fail")

5) Important points to remember

- OR returns TRUE if **any** condition is TRUE
- AND returns TRUE only if **all** conditions are TRUE
- Can handle numbers, text, dates, and logical expressions
- Often used with IF, NOT, and other logical functions

6) When NOT to rely only on OR & AND

- For complex conditional logic involving multiple OR/AND combinations, formulas can become long → consider helper columns or IFS
- Errors occur if conditions are not properly structured

VLookup

1) What is VLOOKUP?

- VLOOKUP is an Excel formula used to search for a value in the first column of a table and return a corresponding value from another column in the same row
- Syntax: =VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])

2) Why we use VLOOKUP?

- To quickly retrieve information from large datasets
- Useful in reports, dashboards, and data analysis
- Saves time compared to manual searching

3) Practical uses of VLOOKUP

- Find employee name from Employee ID → =VLOOKUP(101,A2:C100,2,FALSE)
- Retrieve product price from product list → =VLOOKUP("Apple",A2:D50,3,FALSE)
- Match data across sheets for reconciliation

4) How to use VLOOKUP

- Example 1: =VLOOKUP(101,A2:C10,2,FALSE) → Looks for 101 in first column and returns value from 2nd column
- Example 2: =VLOOKUP("Pen",A2:D50,4,TRUE) → Approximate match, returns 4th column value
- Notes on range_lookup:
 - FALSE → Exact match (recommended)
 - TRUE → Approximate match (table must be sorted)

5) Important points to remember

- Lookup value must be in the first column of table_array

- Exact match is preferred to avoid errors
- Can return numbers, text, or dates
- If value not found, returns #N/A (can be handled with IFERROR)

6) When NOT to rely only on VLOOKUP

- Cannot lookup values to the left of the first column
- For multiple criteria, use INDEX-MATCH or XLOOKUP (newer Excel versions)

HLookup

1) What is HLOOKUP?

- HLOOKUP is an Excel formula used to search for a value in the first row of a table and return a corresponding value from another row in the same column
- Syntax: =HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])

2) Why we use HLOOKUP?

- To quickly retrieve information from datasets arranged in rows
- Useful in reports, dashboards, and horizontal tables
- Saves time compared to manual searching

3) Practical uses of HLOOKUP

- Find monthly sales for a product → =HLOOKUP("Jan",A1:M3,2,FALSE)
- Retrieve data from horizontally structured tables
- Match and extract values across sheets or reports

4) How to use HLOOKUP

- Example 1: =HLOOKUP("ProductA",A1:D5,3,FALSE) → Looks for "ProductA" in first row and returns value from 3rd row
- Example 2: =HLOOKUP(101,A1:Z10,4,TRUE) → Approximate match, returns value from 4th row
- Notes on range_lookup:
 - FALSE → Exact match (recommended)
 - TRUE → Approximate match (row must be sorted)

5) Important points to remember

- Lookup value must be in the first row of table_array
- Exact match is preferred to avoid errors
- Can return numbers, text, or dates
- If value not found, returns #N/A (can be handled with IFERROR)

6) When NOT to rely only on HLOOKUP

- Cannot lookup values above the first row
- For multiple criteria, use INDEX-MATCH or XLOOKUP (newer Excel versions)

XLookup

1) What is XLOOKUP?

- XLOOKUP is an Excel formula used to search a value in a range or array and return a corresponding value from another range, replacing VLOOKUP/HLOOKUP
- Syntax: =XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], [search_mode])

2) Why we use XLOOKUP?

- To look up values both vertically and horizontally
- Can return exact or approximate matches
- Handles missing values with custom output
- More flexible and powerful than VLOOKUP/HLOOKUP

3) Practical uses of XLOOKUP

- Find employee name from ID → =XLOOKUP(101,A2:A100,B2:B100,"Not Found")
- Retrieve product price → =XLOOKUP("Apple",A2:A50,C2:C50,0)
- Replace VLOOKUP in dynamic dashboards for exact or approximate matching

4) How to use XLOOKUP

- Example 1: =XLOOKUP(101,A2:A10,B2:B10,"Not Found") → Returns value from B2:B10 for ID 101
- Example 2 (approximate match): =XLOOKUP(55,A2:A10,B2:B10,,1) → Returns closest larger value
- Can also search from bottom to top using search_mode argument

5) Important points to remember

- Can lookup values to the left or above, unlike VLOOKUP/HLOOKUP
- Default match is exact match
- Can handle errors using if_not_found argument
- Supports dynamic arrays in newer Excel versions

6) When NOT to rely only on XLOOKUP

- Only available in newer Excel versions (Excel 365 & 2021+)
- For extremely large datasets, consider INDEX-MATCH for performance in some cases

Index and Match

1) What is INDEX?

- INDEX returns the value of a cell at a specified row and column in a range or array
- Syntax: =INDEX(array, row_num, [column_num])

2) What is MATCH?

- MATCH returns the relative position of a value in a range
- Syntax: =MATCH(lookup_value, lookup_array, [match_type])

3) Why we use INDEX & MATCH together?

- To perform flexible lookups vertically or horizontally
- Can replace VLOOKUP/HLOOKUP with more control
- Allows lookup to the left, unlike VLOOKUP

4) Practical uses of INDEX & MATCH

- Find employee name from ID:

=INDEX(B2:B100, MATCH(101, A2:A100, 0)) → Returns the value from B2:B100 where ID = 101

- Retrieve product price dynamically →

=INDEX(C2:C50, MATCH("Apple", A2:A50, 0))

- Perform two-way lookup (row & column) for dynamic dashboards

5) How to use INDEX & MATCH

- Example 1: =INDEX(B2:B10, MATCH(55, A2:A10, 0)) → Returns value in B2:B10 corresponding to 55 in A2:A10

- Example 2 (two-way lookup):

=INDEX(B2:E10, MATCH("ProductA", A2:A10, 0), MATCH("Jan", B1:E1, 0)) → Returns value at intersection of ProductA row and Jan column

6) Important points to remember

- MATCH type = 0 → Exact match, 1 → Less than, -1 → Greater than
- Can lookup values anywhere, left, right, top, or bottom
- More efficient for large datasets compared to multiple nested VLOOKUPS

7) When NOT to rely only on INDEX & MATCH

- Slightly more complex syntax for beginners
- For very small datasets, VLOOKUP/HLOOKUP may be simpler to use

Row and Rows

1) What is ROW?

- ROW is an Excel formula used to return the row number of a reference
- Syntax: =ROW([reference])

2) What is ROWS?

- ROWS is an Excel formula used to return the total number of rows in a range
- Syntax: =ROWS(array)

3) Why we use ROW & ROWS?

- ROW → To find the position of a cell in a worksheet
- ROWS → To count total rows in a dataset or range
- Useful in dynamic formulas, data analysis, and array calculations

4) Practical uses of ROW & ROWS

- ROW Example: =ROW(A5) → Returns 5
- ROW Example with range: =ROW(B2:C4) → Returns the first row number of the range (2)
- ROWS Example: =ROWS(A1:A10) → Returns 10 → total rows in the range
- ROWS used in dynamic formulas → e.g., =INDEX(A:A, ROWS(\$A\$1:A1)) for sequential listing

5) Important points to remember

- ROW without reference → returns the row number of the cell where formula is placed

- ROWS always counts number of rows in the given range
- Both can be combined with INDEX, OFFSET, and other dynamic formulas

6) When NOT to rely only on ROW & ROWS

- ROW → Only returns row number, not cell value
- ROWS → Only counts rows, not columns; use COLUMNS() for column count

Column and Columns

1) What is COLUMN?

- COLUMN is an Excel formula used to return the column number of a reference
- Syntax: =COLUMN([reference])

2) What is COLUMNS?

- COLUMNS is an Excel formula used to return the total number of columns in a range
- Syntax: =COLUMNS(array)

3) Why we use COLUMN & COLUMNS?

- COLUMN → To find the position of a column in a worksheet
- COLUMNS → To count total columns in a dataset or range
- Useful in dynamic formulas, data analysis, and array calculations

4) Practical uses of COLUMN & COLUMNS

- COLUMN Example: =COLUMN(B5) → Returns 2 → Column number of B
- COLUMN Example without reference: =COLUMN() → Returns column number of the cell with formula
- COLUMNS Example: =COLUMNS(A1:D10) → Returns 4 → Total columns in the range
- COLUMNS used in dynamic formulas → e.g., =INDEX(A1:D10,1,COLUMNS(\$A\$1:B1)) for sequential listing

5) Important points to remember

- COLUMN without reference → returns column number of current cell
- COLUMNS always counts number of columns in the given range
- Both can be combined with INDEX, OFFSET, and other dynamic formulas

6) When NOT to rely only on COLUMN & COLUMNS

- COLUMN → Only returns column number, not column letter
- COLUMNS → Only counts columns, not rows; use ROWS() for row count

PMT

1) What is PMT?

- PMT is an Excel financial formula used to calculate the payment for a loan or investment based on constant interest rate, number of periods, and present value
- Syntax: =PMT(rate, nper, pv, [fv], [type])

2) Why we use PMT?

- To calculate EMI or periodic payments for loans
- Useful in personal finance, business planning, and budgeting
- Helps in planning repayment schedules

3) Practical uses of PMT

- Calculate monthly EMI $\rightarrow =\text{PMT}(6\%/12, 12*5, -500000) \rightarrow$ 5-year loan at 6% annual rate, PV ₹5,00,000
- Determine periodic investment payment $\rightarrow =\text{PMT}(8\%/12, 12*10, -100000) \rightarrow$ 10-year investment
- Budgeting loan or rent payments based on interest and tenure

4) How to use PMT

- rate $\rightarrow$ interest rate per period
 - nper $\rightarrow$ total number of periods
 - pv $\rightarrow$ present value or principal (use negative value to get positive payment)
 - fv $\rightarrow$ future value (optional, default = 0)
 - type $\rightarrow$ 0 = payment at period end, 1 = payment at period start (optional)
-
- Example 1: $=\text{PMT}(0.05/12, 12*3, -100000) \rightarrow$ Monthly payment for 3-year loan at 5%
 - Example 2: $=\text{PMT}(7\%/4, 4*5, -200000, 0, 1) \rightarrow$ Quarterly payment, 5-year loan, payment at period start

5) Important points to remember

- Present value (pv) should be negative for outgoing payments
- PMT returns amount per period, including principal + interest
- Can be combined with other financial formulas like FV, RATE, NPER

6) When NOT to rely only on PMT

- For loans with variable interest rates, PMT gives approximate value
- Does not separate principal and interest components; use PPMT & IPMT for that

IPMT

1) What is IPMT?

- IPMT is an Excel financial formula used to calculate the interest portion of a payment for a specific period of a loan or investment
- Syntax: $=\text{IPMT}(\text{rate}, \text{per}, \text{nper}, \text{pv}, [\text{fv}], [\text{type}])$

2) Why we use IPMT?

- To find out how much of a payment goes towards interest in a loan or investment
- Useful in loan amortization schedules, budgeting, and financial analysis
- Helps separate principal and interest components of payments

3) Practical uses of IPMT

- Interest on monthly EMI $\rightarrow =\text{IPMT}(6\%/12, 1, 12*5, -500000) \rightarrow$ Interest for first month of 5-year loan
- Interest calculation for quarterly payment $\rightarrow =\text{IPMT}(8\%/4, 2, 4*3, -200000) \rightarrow$ Interest for 2nd quarter of 3-

year loan

- Useful in creating amortization tables for loans

4) How to use IPMT

- rate → interest rate per period
 - per → period for which interest is calculated
 - nper → total number of periods
 - pv → present value or principal (negative for outgoing payment)
 - fv → future value (optional, default = 0)
 - type → 0 = payment at period end, 1 = payment at period start (optional)
-
- Example 1: =IPMT(0.05/12, 3, 12*3, -100000) → Interest for 3rd month of 3-year loan at 5%
 - Example 2: =IPMT(7%/4, 5, 4*5, -200000, 0, 1) → Interest for 5th quarter, payment at period start

5) Important points to remember

- Returns only the interest portion of the payment
- Principal portion can be calculated using PPMT
- Present value (pv) should be negative for outgoing payments

6) When NOT to rely only on IPMT

- For loans with variable interest rates, results are approximate
- Does not calculate total payment; combine with PMT for full EMI

NPV

1) What is NPV?

- NPV (Net Present Value) is used to calculate the present value of future cash flows after deducting the initial investment
- It helps in evaluating whether an investment or project is profitable or not
- Syntax: =NPV(rate, value1, [value2], ...)

2) Why we use NPV?

- To check profitability of projects
- To compare multiple investment options
- Widely used in capital budgeting and financial decision-making

3) Practical uses of NPV

- Project evaluation in companies
- Investment appraisal (machines, plants, new projects)
- Decision making: accept or reject a project

4) How to use NPV (Important concept)

- NPV function in Excel calculates **present value of future cash inflows only**
- Initial investment (cash outflow) is **added separately**

- Example:

Initial Investment = ₹1,00,000

Annual Cash Inflows = ₹30,000 for 5 years

Discount Rate = 10%

Formula:

=NPV(10%, 30000,30000,30000,30000,30000) - 100000

5) Interpretation of NPV result

- NPV > 0 → Project is profitable (Accept)
- NPV = 0 → No gain, no loss
- NPV < 0 → Project is not profitable (Reject)

6) Important points to remember

- Cash flows should be entered in correct time order
- Discount rate should match the cash flow period
- Initial investment should not be included inside NPV values

7) Limitations of NPV

- Assumes constant discount rate
- Results may change with small changes in rate
- Does not show percentage return (use IRR for that)

IRR

1) What is IRR?

- IRR (Internal Rate of Return) is the rate at which Net Present Value (NPV) becomes zero
- It shows the percentage return earned by a project or investment
- Syntax: =IRR(values, [guess])

2) Why we use IRR?

- To measure profitability in percentage terms
- Easy comparison between different projects
- Widely used in capital budgeting decisions

3) Practical uses of IRR

- Project evaluation and selection
- Comparing investment opportunities
- Used by finance professionals and management

4) How to use IRR in Excel

- Cash flows must include **initial investment (negative value)** and future inflows (positive values)
- Cash flows should be in correct time sequence

- Example:

Initial Investment = -₹1,00,000

Cash Inflows = ₹30,000 for 5 years

Formula:

=IRR(-100000,30000,30000,30000,30000,30000)

5) Interpretation of IRR result

- IRR > Cost of Capital → Accept the project
- IRR = Cost of Capital → Indifferent decision
- IRR < Cost of Capital → Reject the project

6) Difference between NPV and IRR

- NPV shows value in absolute amount
- IRR shows return in percentage
- NPV is more reliable for mutually exclusive projects

7) Important points to remember

- Cash flows must have at least one negative and one positive value
- IRR may give multiple results for irregular cash flows
- Use XIRR when cash flows are at irregular dates

8) Limitations of IRR

- Assumes reinvestment at IRR
- May not work properly for non-conventional cash flows
- Can be misleading when comparing different-sized projects

"Next, we will understand different types of charts in Excel and their practical uses for data analysis and reporting."

Column Chart

Definition

- Column Chart displays data using vertical bars
- It is used to compare values across different categories

Uses

- To compare monthly or yearly data
- To show differences between categories clearly
- Commonly used in reports and dashboards

Example

- Comparing monthly sales of different months
- Showing department-wise revenue
- Displaying year-wise profit comparison

Line Chart

Definition

- Line Chart displays data points connected by lines
- It is mainly used to show trends over a period of time

Uses

- To analyze trends and changes over time
- To track growth or decline in data
- Commonly used for time-based data

Example

- Monthly sales trend of a company
- Year-wise profit growth
- Tracking temperature or stock price movement

Pie Chart

Definition

- Pie Chart shows data in a circular form
- It represents part-to-whole relationship using slices

Uses

- To show percentage or proportion of total
- Best when data categories are limited
- Easy to understand for summary reports

Example

- Expense breakup of a company
- Market share of different products
- Percentage-wise distribution of sales

Bar Chart

Definition

- Bar Chart displays data using horizontal bars
- It is used to compare values across categories

Uses

- Useful when category names are long
- To compare data across multiple categories
- Makes comparison easy and clear

Example

- Comparing sales of different products
- Department-wise performance comparison
- Category-wise expense analysis

Area chart

Definition

- Area Chart displays data using lines with the area filled below them
- It is used to show trends along with magnitude of data over time

Uses

- To analyze trend and total value together
- To compare multiple data series over time
- Useful for cumulative data representation

Example

- Year-wise sales growth
- Monthly revenue trend
- Cumulative production output over time

Scatter Chart

Definition

- Scatter Chart displays data points on X and Y axes
- It is used to show the relationship between two numeric variables

Uses

- To identify correlation between two variables
- To analyze patterns and trends in data
- Commonly used in statistical analysis

Example

- Relationship between sales and advertising cost
- Comparison of height and weight data
- Analysis of marks vs study hours

Map Chart

Definition

- Map Chart displays data values on a geographical map
- It is used to show location-based data visually

Uses

- To compare data across regions or locations
- To analyze geographical distribution of data
- Useful for regional performance analysis

Example

- State-wise sales performance
- Country-wise revenue comparison
- Region-wise population data

Stock Chart

Definition

- Stock Chart displays financial data such as Open, High, Low, and Close prices
- It is mainly used for analyzing stock market trends

Uses

- To track price movement of stocks
- To compare stock performance over time
- Useful for investors and financial analysts

Example

- Daily stock prices of a company
- Comparison of multiple stocks' high and low values
- Analyzing weekly or monthly stock trends

Surface Chart

Definition

- Surface Chart displays data across a 3D surface
- It is used to find optimum combinations between two sets of data

Uses

- To visualize data patterns in three dimensions
- To identify peaks and valleys in data
- Useful for analysis of complex datasets

Example

- Profit variations based on price and demand
- Sales performance across regions and months
- Temperature distribution across a geographical area

Bubble Chart

Definition

- Bubble Chart is a type of Scatter Chart where a third value is represented by the size of the bubble
- It is used to show relationship between three numeric variables

Uses

- To analyze three-dimensional data visually
- To compare data points with size and position
- Useful for business and financial analysis

Example

- Sales vs Advertising Cost vs Profit (bubble size = Profit)
- Population vs GDP vs Area (bubble size = Area)
- Marketing campaign performance with reach, engagement, and budget

Radar Chart

Definition

- Radar Chart displays data values on axes starting from the same point
- It is used to compare multiple variables for one or more items

Uses

- To visualize performance across multiple parameters
- Useful in skill assessment, scorecards, and benchmarking
- Helps in identifying strengths and weaknesses

Example

- Employee skill assessment across 5 skills
- Department performance comparison
- Product feature comparison

Treemap Chart

Definition

- Treemap Chart displays hierarchical data as nested rectangles
- Size and color of each rectangle represent the value

Uses

- To show part-to-whole relationships in a hierarchy
- Useful for visualizing large datasets
- Helps in spotting patterns and outliers quickly

Example

- Sales by category and subcategory

- Budget allocation across departments
- Website traffic by section and page

Sunburst Chart

Definition

- Sunburst Chart shows data in rings to represent levels of hierarchy
- Each ring shows a category and its subcategories

Uses

- To visualize multi-level data clearly
- To show how parts make up a whole
- Easy to see relationships between main and subcategories

Example

- Company sales → Region → State → City
- Product sales → Category → Subcategory → Product
- Budget allocation → Department → Team → Project

Histogram Chart

Definition

- Histogram Chart shows the distribution of numeric data into bins or intervals
- It is used to see how often values occur within a range

Uses

- To analyze frequency of data points
- To understand data distribution and patterns
- Commonly used in statistics and data analysis

Example

- Marks distribution of students in a class
- Monthly sales range analysis
- Age group distribution of customers

Waterfall Chart

Definition

- Waterfall Chart shows how an initial value increases or decreases through a series of positive and negative values
- It is used to visualize cumulative effect on a value

Uses

- To track profit and loss over time
- To show how different factors affect a total
- Useful in financial analysis and reporting

Example

- Company starting revenue → costs → profit
- Monthly income and expense impact on net cash
- Step-by-step changes in project budget

Funnel Chart

Definition

- Funnel Chart shows data in stages, decreasing in size like a funnel
- It is used to visualize a process or flow with progressively smaller values

Uses

- To analyze sales or conversion process
- To identify drop-offs in a process
- Useful for business and marketing analysis

Example

- Lead → Contacted → Proposal → Closed Deal
- Website visitors → Sign-ups → Purchases
- Recruitment process: Applicants → Interviews → Hired

Combo Chart

Definition

- Combo Chart combines two or more chart types in a single chart
- It is used to show different types of data together for better comparison

Uses

- To compare data series with different scales
- To show trends and values simultaneously
- Useful for financial and performance analysis

Example

- Column for sales and line for profit margin
- Bar for expenses and line for revenue
- Comparing target vs actual performance

Doughnut Chart

Definition

- Doughnut Chart is similar to a Pie Chart but with a hole in the center
- It is used to show part-to-whole relationships with multiple series

Uses

- To compare multiple data series
- To visualize proportions clearly
- Useful for dashboards and reports

Example

- Sales distribution by product category and subcategory
- Expense breakdown for multiple departments
- Revenue contribution by region and product

"Next, we will understand the different tabs available in Excel and their important options used in daily work."

Home Tab

Purpose

- Home Tab is used for basic data entry, formatting, editing, and managing worksheet content
- Most daily Excel work is performed from this tab

Major groups & options in Home Tab

1) Clipboard

- Paste – Paste copied or cut data
- Cut – Move data from one place to another
- Copy – Copy data without removing it
- Format Painter – Copy formatting from one cell to another

2) Font

- Font Type – Change text style
- Font Size – Change text size
- Bold, Italic, Underline – Highlight text
- Borders – Apply cell borders
- Fill Color – Add background color to cells
- Font Color – Change text color
- More Font Options – Strikethrough, Subscript, Superscript, Effects

3) Alignment

- Horizontal Alignment – Left, Center, Right
- Vertical Alignment – Top, Middle, Bottom
- Wrap Text – Display long text within a cell
- Merge & Center – Merge multiple cells and center text

- Orientation – Rotate text vertically, diagonally, or stacked
- Increase / Decrease Indent – Adjust spacing from cell border

4) Number

- Number Formats – General, Number, Currency, Accounting, Date, Time, Percentage, Fraction, Scientific, Text
- Comma Style – Add thousand separator
- Increase / Decrease Decimal – Adjust decimal places
- Accounting Number Format – Format currency with alignment

5) Styles

- Conditional Formatting – Highlight cells based on rules
- Format as Table – Convert data into a structured table with styles and filters
- Cell Styles – Apply predefined styles (Heading, Total, Good, Bad, Input, etc.)

6) Cells

- Insert – Insert new cells, rows, or columns
- Delete – Delete cells, rows, or columns
- Format – Row height, column width, hide/unhide, protection, cell size adjustments

7) Editing

- AutoSum – Quickly calculate Sum, Average, Count, Max, Min
- Fill – Copy data or create series (Fill Down, Fill Right, Series)
- Clear – Clear contents, formats, or comments from selected cells
- Sort & Filter – Sort data A-Z, Z-A, Custom Sort, Apply / Clear Filter
- Find & Select – Find, Replace, Go To, Select Special, Go To Comments / Objects

Practical Use

- Formatting financial statements, reports, and dashboards
- Cleaning and organizing raw data
- Quick calculations and analysis

Insert Tab

Purpose

- Insert Tab is used to add objects, visuals, tables, charts, and tools into a worksheet
- Helps in data analysis, visualization, and presentation

Major groups & options in Insert Tab

1) Tables

- Table – Convert selected data into a structured table with filters and formatting
- PivotTable – Summarize, analyze, and report large datasets
- Recommended PivotTables – Excel suggests PivotTables based on data
- PivotChart – Create chart based on PivotTable

2) Illustrations

- Pictures – Insert images from computer
- Online Pictures – Insert images from web
- Shapes – Insert rectangles, arrows, flowchart symbols, callouts
- Icons – Insert ready-made icons
- 3D Models – Insert 3D objects
- SmartArt – Create process diagrams, organizational charts, and flowcharts
- Screenshot – Insert screenshot or screen clipping

3) Add-ins / Apps

- Get Add-ins – Insert Excel Add-ins to extend functionality
- My Add-ins – View installed Add-ins

4) Charts

- Recommended Charts – Excel suggests best chart type for selected data
- Insert Column or Bar Chart – Clustered, Stacked, 100% Stacked
- Insert Line or Area Chart – Line, Stacked Line, Area, Stacked Area
- Insert Pie or Doughnut Chart – Pie, Doughnut
- Insert Hierarchy Chart – Tree Map, Sunburst
- Insert Statistic Chart – Histogram, Box & Whisker, Pareto
- Insert Scatter (X, Y) or Bubble Chart – Scatter, Bubble
- Insert Combo Chart – Combine two chart types

5) Maps

- Map Chart – Geographic data visualization by country, region, or state

6) Sparklines

- Line – Small line chart in a single cell
- Column – Mini column chart in a cell
- Win/Loss – Show positive vs negative in a cell

7) Filters

- Slicer – Visual filter for Tables and PivotTables
- Timeline – Date-based filter for PivotTables

8) Links

- Hyperlink – Link to website, file, email, or another sheet

9) Text

- Text Box – Insert text anywhere in sheet
- Header & Footer – Add header or footer for printing
- WordArt – Decorative text for titles
- Signature Line – Add digital signature
- Object – Insert external files (Word, PDF, Excel, etc.)
- Equation – Insert mathematical equation
- Symbol – Insert special characters

10) Media (Excel 365 / Online versions)

- Online Video – Insert video from web
- Audio / Sound Clips – Insert audio

Practical Use

- Creating dashboards and reports
- Visualizing data using charts, tables, and Sparklines
- Adding images, diagrams, and objects for presentation
- Filtering data interactively with Slicers and Timelines

Page Layout

Purpose

- Page Layout Tab is used to control how a worksheet looks when printed
- It helps in setting page size, margins, orientation, and print formatting

Major options in Page Layout Tab

1) Themes

- **Themes** – Apply a complete design (colours, fonts, effects) to the workbook
- **Colours** – Change colour scheme
- **Fonts** – Change default font style
- **Effects** – Apply visual effects
- Used to keep a uniform professional look

2) Page Setup

- **Margins** – Set page margins (Normal, Wide, Narrow, Custom)
- **Orientation** – Choose Portrait or Landscape
- **Size** – Select paper size (A4, Letter, etc.)
- **Print Area** – Define specific area to print
- **Breaks** – Insert or remove page breaks
- **Background** – Add background image (not printed)

3) Scale to Fit

- **Width** – Fit sheet to specific number of pages horizontally
- **Height** – Fit sheet to specific number of pages vertically
- **Scale** – Increase or decrease print size
- Used to fit large data on limited pages

4) Sheet Options

- **Gridlines**

View: Show or hide gridlines on screen

Print: Print gridlines

- **Headings**

View: Show row and column headings

Print: Print row and column headings

5) Arrange

- **Bring Forward / Send Backward** – Arrange objects order
- **Align** – Align shapes, charts, pictures
- **Group / Ungroup** – Group multiple objects
- **Rotate** – Rotate objects
- Used mainly when working with charts, shapes, dashboards

Practical Use

- Printing reports, statements, and MIS
- Setting professional page layout
- Fitting data properly on printed pages

Formula Tab

Purpose

- Formulas Tab is used to work with formulas and functions
- It helps in calculations, data analysis, and managing named ranges

Major options in Formulas Tab

1) Function Library

- **Insert Function** – Helps find and insert the correct function
- **AutoSum** – Quickly adds Sum, Average, Count, Max, Min
- **Financial** – Functions like PMT, IPMT, NPV, IRR
- **Logical** – IF, AND, OR, IFERROR
- **Text** – LEFT, RIGHT, MID, LEN, TRIM, UPPER, LOWER
- **Date & Time** – TODAY, NOW, DATEDIF, EDATE, EOMONTH
- **Lookup & Reference** – VLOOKUP, HLOOKUP, XLOOKUP, INDEX, MATCH
- **Math & Trig** – SUM, ROUND, ABS, MOD
- **More Functions** – Statistical, Engineering, Cube, Information

2) Defined Names

- **Name Manager** – View, edit, or delete named ranges
- **Define Name** – Assign a name to a cell or range
- **Use in Formula** – Insert named ranges into formulas
- Used to make formulas easier to understand and manage

3) Formula Auditing

- **Trace Precedents** – Shows cells used in a formula
- **Trace Dependents** – Shows where a formula is used
- **Remove Arrows** – Remove tracing arrows
- **Show Formulas** – Display formulas instead of results

- **Error Checking** – Find formula errors
- **Evaluate Formula** – Check calculation step by step
- **Watch Window** – Monitor selected cells

4) Calculation

- **Calculation Options** – Automatic / Manual calculation
- **Calculate Now** – Recalculate entire workbook
- **Calculate Sheet** – Recalculate current worksheet

Practical Use

- Writing and managing formulas
- Debugging and checking errors
- Working with complex calculations efficiently

Data Tab

Purpose

- Data Tab is used to import, clean, validate, organize, and analyze data
- Most data-handling and analysis tools are available in this tab

Major options in Data Tab

1) Get & Transform Data (Power Query)

- Get Data – Import data from Excel, Text/CSV, Web, Database, etc.
- Recent Sources – View recently used data sources
- Existing Connections – Use already saved connections

2) Queries & Connections

- Queries & Connections – View and manage imported queries
- Refresh and control data connections

3) Sort & Filter

- Sort A to Z / Z to A – Arrange data
- Sort – Custom sort with multiple levels
- Filter – Apply filters to data
- Clear – Remove applied filters
- Reapply – Reapply filters after data changes

4) Data Tools

- Text to Columns – Split text into multiple columns
- Flash Fill – Automatically fill data based on pattern
- Remove Duplicates – Delete duplicate records
- Data Validation – Restrict data entry (dropdowns, limits)
- Consolidate – Combine data from multiple ranges
- Relationships – Create relationships between tables

5) Forecast

- What-If Analysis – Goal Seek, Scenario Manager, Data Table
- Forecast Sheet – Create future trend forecasts

6) Outline

- Group – Group rows or columns
- Ungroup – Remove grouping
- Subtotal – Insert automatic subtotals

Practical Use

- Importing and cleaning raw data
- Validating and organizing data
- Performing analysis and forecasting

Review Tab

Purpose

- Review Tab is used for checking, protecting, and collaborating on a workbook
- Helps in proofreading, adding comments, and securing worksheets

Major options in Review Tab

1) Proofing

- **Spelling** – Check spelling in the worksheet
- **Research** – Look up definitions or references (depends on Excel version)
- **Thesaurus** – Find synonyms for selected words

2) Accessibility

- **Check Accessibility** – Ensure workbook content is accessible to everyone

3) Insights / Smart Lookup

- **Smart Lookup** – Search the web for information about selected text
- **Insights** – Provides additional information or definitions

4) Language

- **Translate** – Translate selected text into another language
- **Language Preferences** – Set editing and display languages

5) Comments

- **New Comment / Insert Comment** – Add comments to a cell
- **Delete Comment** – Remove comments
- **Show / Hide Comments** – Display or hide comments
- **Previous / Next** – Navigate between comments

6) Notes (in newer Excel versions)

- **New Note** – Add a note (smaller, simpler than comment)

- **Delete Note** – Remove notes
- **Show / Hide Notes** – Display or hide notes

7) Protect

- **Protect Sheet** – Restrict editing for specific cells or the whole sheet
- **Protect Workbook** – Restrict changes to workbook structure
- **Allow Users to Edit Ranges** – Let certain users edit specific ranges

Practical Use

- Reviewing data for errors
- Collaborating with colleagues using comments or notes
- Protecting worksheets to avoid accidental changes

View Tab

Purpose

- View Tab is used to control how the worksheet is displayed on screen
- Helps in managing workbook views, showing/hiding elements, and window arrangement

Major options in View Tab

1) Workbook Views

- **Normal** – Default worksheet view
- **Page Break Preview** – See where pages will break when printed
- **Page Layout** – View worksheet as it will appear on printed page
- **Custom Views** – Save specific view settings for later use

2) Show

- **Ruler** – Display rulers for alignment
- **Gridlines** – Show or hide gridlines on screen
- **Formula Bar** – Show or hide the formula bar
- **Headings** – Show or hide row and column headings

3) Zoom

- **Zoom** – Increase or decrease worksheet view
- **100%** – Reset zoom to 100%
- **Zoom to Selection** – Zoom in on selected cells

4) Window

- **New Window** – Open same workbook in a new window
- **Arrange All** – View multiple windows at the same time
- **Freeze Panes** – Keep specific rows/columns visible while scrolling
- **Split** – Split the worksheet into multiple panes
- **Hide / Unhide** – Hide or show workbook windows

5) Macros

- **Record Macro** – Record a series of actions
- **Use Relative References** – Record macros using relative cell references
- **Macros** – View, edit, or run existing macros

Practical Use

- Adjusting worksheet view for easier data entry or analysis
- Comparing multiple sections or sheets
- Freezing headers or columns while scrolling
- Recording repetitive tasks using Macros